

STOCK MARKET ANALYSIS

USING PYTHON



GETTING THE STOCK MARKET DATA & VISUALISATION

- Choose the stocks you want to analyse
- Download the historical data of the stock using yfinance
- Check the data and give your views about the data's quality

1 - Fetch the data with different interval and period

It takes the following parameters:

- 1.ticker: The name of the tickers you want the stock market data for. If you want stock market data for multiple tickers then separate them by space
- 2.period: The number of days/month of stock market data required. The valid frequencies are 1d, 5d, 1mo, 3mo, 6mo, 1y, 2y, 5y, 10y, ytd, max
- 3.interval: The frequency of the stock market data. The valid intervals are 1m, 2m, 5m, 15m, 30m, 60m, 90m, 1h, 1d, 5d, 1wk, 1mo, 3mo

2 - Resample Stock Data

Convert 1-minute data to 1-hour data or Resample Stock Data

During strategy modelling, you might be required to work with a custom frequency of stock market data such as 15 minutes or 1 hour or even 1 month.

If you have minute level data, then you can easily construct the 15 minutes, 1 hour or daily candles by resampling them. Thus, you don't have to buy them separately.

In this case, you can use the pandas **resample** method to convert the stock market data to the frequency of your choice. The implementation of these is shown below where 1-minute frequency data is converted to 10-minute frequency data.

3 - Visualizing the Stock Data using matplotlib

- To visualize the adjusted close price data, you can use the matplotlib library and plot method.
- improve the plot by resizing, giving appropriate labels and adding grid lines for better readability.

Why use yfinance library?

- 1.Adjusted close price stock market data is available
- 2.The most recent stock market data is available
- 3.Doesn't require an API key to fetch the stock market data

GETTING THE FUNDAMENTAL DATA USING YFINANCE

Fetch the Key Ratios

You can fetch the latest price to book ratio and price to earnings ratio

Revenues

fetching the revenues for ex

```
#revenue = msft.financials.loc['Total Revenue']
```

Note - plot bar graph for the data along with labels

Earnings Before Interest and Taxes (EBIT)

```
EBIT = msft.financials.loc['Earnings Before Interest and Taxes']
```

Note - plot bar graph for the data along with labels

Balance sheet, cash flows and other information

```
# show income statement
```

```
msft.financials
```

```
# show balance sheet
```

```
msft.balance_sheet
```

```
# show cashflow
```

```
msft.cashflow
```

```
# show other info
```

```
msft.info
```